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CMP6200/DIG6200

Individual Undergraduate Project (FYP)

Final Report

**The application for learning Chinese and the
other languages**

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Abstract

As technology progresses at an ever-increasing speed with the emergence of artificial intelligence and improved capabilities of mobile devices, language acquisition is beginning to evolve from traditional grammar and vocabulary tools toward more contemporary models such as real-time tools and interactive methods for individual study. Speaking Chinese is particularly difficult due to the nature of pronunciation (tones); therefore, users learning Chinese continue to identify opportunities for improvement in many free or beginning application-based learning platforms that typically do not provide satisfactory pronunciation feedback. A large percentage of these applications rely upon providing users with basic feedback; indicating whether a user's response was correct but providing little if any detailed guidance regarding the nature of the error (characteristics of letters). To address this problem, this project developed and implemented an offline, mobile-based speech recognition supported tool to provide Chinese learners with a low-latency and locally executed/possibly extensible pronunciation tool for use while learning Chinese or other languages.

A development-based research method was used throughout this project in conjunction with system prototyping. The system that was produced is based on local offline speech recognition technologies for use with mobile learning (i.e., m-learning) applications. Although the initial conception of the system included a combination of speech recognition, tone analysis, and graphical visual feedback to provide a more comprehensive pronunciation-learning experience, practical considerations during development resulted in a gradual shift from an initial focus on creating a combination of all three technologies as independent modules to developing and integrating the offline speech recognition module (OSRM) that uses the core whisper.cpp speech recognition engine and loads ggml-format models locally for use as inference engines. The core functions that were implemented/tested in the final prototype include: loading of pre-trained models into memory; controlling the parameters for performing inference; producing basic speech-to-text conversions, interacting with the user through a graphical user interface (GUI). In addition to being developed on Windows, a basic functional version of the prototype for Android was also developed and evaluated to demonstrate that the operations required for performing local offline speech recognition may be used in the targeted mobile learning application.

The development process revolved around solving practical engineering issues like `std::clamp` computed errors resulting from differing compiler standards clashing, Windows path reading issues, GPU backend configuration issues and dependency linking issues. Continually debugging and refactoring the structure of the project resulted in a working prototype capable of stably completing phrase level Chinese speech recognition and proved out the feasibility of an offline speech recognition method on personal devices. Offline prototype does deliver on privacy, independence and extensibility while requiring further Android optimisation, detailed pronunciation feedback, user testing and educational evaluation.

Keywords: computer-assisted language learning; offline speech recognition; whisper.cpp; Chinese learning; pronunciation feedback; mobile learning

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1.0 Introduction and Context

The evolution of artificial intelligence, mobile Internet, and terminal computing power has resulted in a shift from traditional classroom-based and static methods of teaching towards more intelligent, interactive, and customisable forms of digital language acquisition. Now, ASR and CALL/CAL provide new opportunities for learners to develop their listening and speaking skills. One of the major obstacles to Chinese as a second language learners is the fact that pronunciation can be difficult when speaking a tonal language such as Mandarin, as it has four lexical tones, as well as a neutral tone in some words and contexts that can all have an impact on the meaning of the spoken word. While learners may acquire some basic vocabulary or grammatically correct forms reasonably quickly, they often have difficulty recognising and producing tonal distinctions correctly. This can impact their ability to successfully communicate with others and the effectiveness of any automated language learning material that they may be using for both formal and informal study.

Research reviewed during the interim stage of this project suggests that the inclusion of visual information can improve the perception of Mandarin tones, with particularly noticeable benefits for beginner learners (Farran & Morett, 2024; Han, et al., 2019; Garg, et al., 2023).

At the same time, while many language learning applications have moved towards relative maturity in areas like vocabulary memorisation, grammar training, and listening practice, support for pronunciation is still rather basic. In many cases, learners are given only a simple “correct” or “incorrect” judgement, without being shown where the error occurs or how it can be corrected. As a result, even when learners realise that their pronunciation is inaccurate, they may still struggle to identify the specific problem independently, which can negatively affect both confidence and learning motivation.

Against this background, the project focuses on developing an offline Chinese speech recognition prototype as a technical foundation for future pronunciation-feedback functions.

1.1 Problem statement

The above project seeks to solve the problem that existing pronunciation-training applications for the Chinese language are limited in their feedback granularity, interpretability, privacy protection, and offline usability. These limitations create a lack of timely, effective, and sustainable support for beginners correcting their pronunciation errors when learning Chinese. In independent practice scenarios in particular, learners frequently receive only simple “correct” or “incorrect” feedback, without being shown where the error occurs or how it can be improved. (Han, et al., 2019; Garg, et al., 2023)

The importance of addressing this issue is amplified by its role within self-driven language learning. In the absence of clear and meaningful feedback, learners may notice that their pronunciation is wrong but will still have difficulty figuring out how to fix the problem or where to adjust. As a result, after a while, their confidence will decrease as well as their motivation to continue learning and producing new sounds. Also, many current solutions require a large amount of cloud computing resources which create new restrictions on latency in responding, privacy of data, and long-term access to the data. For pronunciation learning applications, where repeated and immediate practice is important, such limitations reduce practical usefulness, particularly for users who wish to practice in personal and low-resource environments (He, et al., 2019; Jacob, et al., 2018).

Therefore, this project aims to create a local offline prototype which will provide the foundation for future pronunciation-learning support that is both more private and technically independent. At present, this project has primarily been focused on establishing the feasibility of an offline Chinese speech recognition system rather than an actual full pronunciation-feedback system; nevertheless, the work completed to this point has directly addressed the limitations noted above, and has laid the groundwork for developing a more detailed version of the feedback functionality during subsequent development stages.

2.0 Review of Existing Knowledge

First, within the literature related to the learning of Mandarin as a second language by adult English speakers, tone (also called pitch or melody) has traditionally been viewed as one of the most difficult elements to learn. Studies have shown that visual aids may promote the perception of Mandarin tone and facilitate learning at specific levels of language proficiency, especially with young

learners; however, the level of improvement based on these previous studies is dependent upon the type of task and the level of proficiency of the participants as well as the type of visual aid. Collectively, this literature suggests that by using visual cues to help learners perceive tonal distinctions and remember how to pronounce words with tonal patterns, learners' success in developing pronunciation skills will be greater. In the context of this project, these findings support the view that combining auditory and visual input is an important direction in second-language pronunciation teaching (Farran & Morett, 2024; Garg, et al., 2023; Han, et al., 2019).

Real-time speech recognition has been made more technically possible on mobile devices. Studies have demonstrated that many modern end-to-end speech recognition models can process data in real-time on mobile devices through streaming implementations and that quantising techniques can decrease both latency and power however they will not cause a significant decrease in accuracy. These developments provide an important theoretical basis for the project's preference for a lightweight, local, and low-latency system design (He, et al., 2019; Jacob, et al., 2018).

Thirdly, the increase in interest in employing visual feedback for pronunciation teaching. Research shows that if students have visual information available to them regarding their own articulation, they can create a spatial awareness of how oral movement, place of articulation, and the production of sound, interact. This type of feedback is more intuitive than a purely abstract verbal explanation so will provide better support for identifying and correcting errors. For this reason, the early design stage of the project originally incorporated facial landmark detection into the system concept, with the aim of using lip and jaw movement data to generate more interpretable pronunciation feedback (Bliss, et al., 2018; Google, 2024). In this part, Google's MediaPipe documentation is used as a technical reference for facial landmark detection rather than as evidence of educational effectiveness.

All of the above studies already point to the relatively clear theoretical and technical direction of "speech recognition + multimodal feedback + mobile learning", but clarify the combinatorial principle here is more difficult, at least in the context of Chinese pronunciation learning, and especially beginners, the combination of local processing, real-time feedback, educational interpretability and deployment feasibility of systematic solutions have yet to be significantly explored. This gap provides the direct research context for the present project.

2.1 Critical Analysis and Gap Identification

Various reports of the results of former research have provided useful information regarding tone perception, visual assistance and mobile speech processing; however, there are still many areas where improvement is needed. For example, most existing systems provide users with only a basic level of feedback with respect to their performance; the most common feedback is through binary classification (i.e., 'correct' or 'incorrect'). As a result, learners are often not told how serious the error is, what type of mistake has been made, or how they might improve it (Han, et al., 2019; Garg, et al., 2023).

In summary, application of this technology in education, pronunciation-training systems, is still limited, although mobile, facial landmark tracking technology is now very advanced and technology to support this area is available and in existence. This creates a gap between technical feasibility and meaningful pedagogical implementation (Google, 2024; Garg, et al., 2023).

In addition, a significant number of systems still rely on cloud-based processing. Cloud-based designs may indeed enable powerful computational capabilities, but they may also introduce issues such as high latency, high energy consumption and privacy concerns for end users. For learning tools that rely on repeated, immediate practice, these factors may reduce usability, especially in mobile environments or personal learning situations where stable connectivity cannot always be assumed (He, et al., 2019; Jacob, et al., 2018).

The previous research that was examined does not adequately address the needs of true beginner learners or the importance of providing supportive environments for self-directed learning. Thus, learners unfamiliar with such studies and/or not entirely conversant in the language they are learning, may struggle to leverage recognition feedback into a practical idea for correcting and improving. Beyond these gaps highlighted in the literature however, the 'real world' use of this project exposed another problem. Although the theory percolating these elements together with speech recognition, pronunciation scoring and visual feedback is exciting, practically a lot of the pain point doesn't come from the actual model but at a more engineering level: questions of compatibility, build management, hardware issues and system integration on real devices during local deployment. A later record of this project shows that compiler standard consistency, reliability of GPU acceleration, dependency linking and general reliability in deployment, turned out to be much harder than expected. Thus, the gap which this project aims to partner with can be summarized as the lack of a beginner friendly local model of a Chinese pronunciation learning system that uses offline speech recognition and builds the foundation for a more practical attention to pronunciation feedback in the future.

2.2 Project Justification

This submission is justified by engineering and educational considerations. Engineering-wise, it tests whether offline Chinese speech recognition can be operationalized in a functional learning prototype on personal devices, whereas from an educational perspective, this prototype does not intend on being limited to a transcription tool, but rather on laying the technical groundwork for later pronunciation-feedback syllable comparison, error identification, similarity scoring.

In terms of “extract of this work used in the wild” edge cases, the value of the project isn’t limited to Chinese language learners. The modelling work could potentially also be extended to other language learning contexts where pronunciation and its computerised feedback service matters. Even to other language areas not primarily based in Chinese learning, the modular structure design may yet conditionally inspire: it wouldn’t translate directly into a pronunciation-support tool, but it might shed new insights on how pronunciation-support tools play as part of a larger training or rehabilitation, for instance in clinical speech pathology contexts - but separate specialist research would happen there (Bliss, et al., 2018; Olson & Offerman, 2021) .

3.0 Project Aims, Objectives, and Scope

This section renders the assessed issue and research gap into a brief project plan, with an expression of the guiding purpose of the project file, particular operational objectives for achieving that purpose and the limits (boundaries) within which we had to operate.

3.1 Project Aim

The aim of this project is to design, implement, and evaluate an offline speech recognition-assisted learning prototype for Chinese and other language-learning scenarios, with a particular focus on supporting pronunciation learning in a low-latency, privacy-conscious, and locally executable manner.

At the current stage, the project places primary emphasis on validating the feasibility of offline Chinese speech recognition on personal devices, while also establishing a technical foundation for future expansion into a more complete pronunciation feedback system.

3.2 Project Objectives

To achieve said goal, the present project will consist of the following objectives:

1. To review existing academic and technical research related to Chinese pronunciation learning, multimodal feedback, and offline speech recognition, to identify a clear research gap and justify the project direction (Han, et al., 2019; Farran & Morett, 2024; He, et al., 2019; Jacob, et al., 2018).
2. to create a skewed prototype of system architecture for localization with respect to local model loading, speech input processing, and output transcription; compatible with future development of pronunciation feedback functionality.
3. to develop an offline Chinese speech recognition prototype using a lightweight localized deployment method, including integration of the models, building of the build configurations, and user interfaces.
4. to evaluate the performance of the prototype with respect to stability when recognizing phrases monthly for a period sufficient to achieve meaning.
5. to critically evaluate the prototype with respect to its strength, weaknesses and potential for future development for the purposes of educational applications, privacy issues, and practical application development.

The objectives reflect the original pedagogical intent of this project's development and the better technical decision in the intervening time as regards offline speech recognition as the feasible building block given the short time frame.

3.3 Project Scope

The scope of this project is defined to keep the work focused, realistic, and aligned with the available time and technical resources.

The project includes the following:

- the design and implementation of an offline speech recognition-assisted learning prototype.
- local loading and inference of a speech recognition model for Chinese input.
- a basic interactive interface supporting model loading, recording control, and transcription display.
- implementation of both a Windows development version and a basic Android version of the prototype.
- preliminary functional prototyping/testing of phrase-level Chinese speech recognition with visual feedback.
- proof of concept for extension into pronunciation feedback and other learning support functions.

The project includes the following at this early stage:

- a fully completed visual feedback system based on facial landmark tracking.

- large user experiments/actual pedagogical studies.
- production quality software for multiple platforms/use in the wild.
- advanced grammar teaching, semantic learning, or continuous long-speech recognition tasks.

This scope reflects the practical decision to prioritise a stable and functional offline prototype first, rather than attempting to complete all originally envisioned features within a single development cycle.

4.0 Project Design

This was intended as a practice-based development study using staged and iterative prototyping. The first stage was used to define the educational problem and to review the literature and a technical feasibility. While the original design called for incorporation of speech recognition and pronunciation analysis alongside visual feedback, the scope was reduced for technical complexity and for time considerations. The nearer goal therefore was a proof-of-concept offline speech recognition prototype, with the feedback and others to follow.

From a project design standpoint this may not be viewed as a weakness, instead it is a real world and iterative example of project management. The development process revealed that deployment, model integration, dependencies and adaptation to hardware was more challenging than initially expected, thus the design strategy evolved—from building the complete multimodal learning system straightaway—to prioritising the most technically basic and achievable component. This insured that a final artefact could be built which was still usable, relevant and evaluable within the project at hand.

The problem is that of the quality of existing pronunciation-learning systems in terms of feedback, privacy and offline usability. The response aimed for is a local offline prototype that can confirm technical feasibility of speech recognition on personal devices and provide a platform for later educational expansion. The design methods are further explained in the next subsection.

4.1 Design and Methods

In line with the Final Year Project requirements, and considering the focus on functional verification, this project can be also considered development based, in that the study is undertaken not through speculating but through building and testing a working system prototype while working through the research questions. The project adopts a user-oriented and iterative approach. The first stage mainly focused on defining precisely the main problems faced by a Chinese learner in getting pronunciation training via revisiting the literature and

the learning applications. This laid a clear (one) educational problem and an (two) technical path. The middle stage sees the main work being done on the technical implementation and full-on integration of the offline speech recognition module. In the last, later stage, the project has begun to move gradually from the existed prototype heading towards pronunciation feedback mechanisms, performance optimisation and testing for actual users. The interim report envisaged a much wider piecing together of speech and visual modules, together with questionnaires and collecting experimental feedback; in practice the project has moved more towards completing the offline speech recognition prototype and likewise creating a basic user interface-a practical foundation for further evaluative as well as additional system movement. The project, methodologically, mainly employs three methods. First, review. Literature was reviewed to form the theoretical basis of the study and to give a clear-cutting point for the study. Revisiting automatic speech recognition, mobile inference, Chinese tone instruction, and visual feedback literature together helped in defining the educational problem more clearly, as well as justified the system overall direction (Han, et al., 2019; Farran & Morett, 2024; He, et al., 2019; Jacob, et al., 2018; Bliss, et al., 2018).

The second method of conducting the system development for the project was a practical implementation of the research ideas. The main objective of the project is to implement offline AI models and create an application prototype with a working system as the main research output. Thus, the project will examine the research ideas through concrete engineering implementation and testing of model integration, build configuration, interface development, and required functional changes.

Thirdly, functioning verification will be conducted to assess the prototype's current state and its development for submission to an overarching research proposal. The preliminary assessment phase will consist of conducting prototype testing, module debugging, and initial testing to determine if the prototype is logical, has stable identification of features, and provides the groundwork for future scalability. Because large-scale user testing has not yet been 100% complete, the results from the initial assessment will largely be limited to prototype verification and initial feasibility assessment, and not as a full report of pedagogy effectiveness.

Due to the nature of this project in the context of the projects' purpose and recent development, all three of the methods utilised within the current framework provided a suitable and reasonable methodology to accomplish an objective and realistic framework for the project.

To assess preliminary learner expectations regarding the pronunciation support functionality of the project, a small-scale survey was administered. Informal criteria for recruitment of participants and timeframe were employed to encourage random responses. Questions in the survey queried the participant's perceived problems with pronunciation, expectation of immediate pronunciation feedback, and their attitudes toward using the functionality in an offline manner. This sample yielded a limited number of responses; therefore, the results of the survey can be considered contextual user verification only, and not a legitimate generalisation of statistical significance.

4.2 Justification of Methods

There are several important considerations that influenced the selection of the above approaches.

In addition to differences in types of research methodology, the development of a prototype system that can be physically built, implemented and executed and is the primary product of this project. Thus, a development-based research strategy is appropriate since it will enable knowledge generation through building, testing and improving a functional artefact instead of using only theoretical models for discussion.

Moreover, several challenging elements of an offline speech recognition system have a strong engineering element to them, with many of the significant challenges such as integrating multiple models, deploying locally, managing the development process, ensuring the hardware will run and maintaining the stability of the final system will not be able to be fully understood solely on a theoretical basis. All will be revealed and addressed only through actual implementation, continuous changes to correct identified issues and finally testing in a real-world development environment. Consequently, implementing the system is not a separate phase of the project; it is an additional integral component of the research process.

The justification for prioritising the local offline prototype was also an education-based one, as later features like pronunciation feedback, learner guidance and teaching evaluation depend on a stable technological base upon which to build those later features. This decision to utilise the combined methodologies of Literature Review, System Development and Functional Verification for this project was supported by three key reasons. Each of the components of the combined methodologies supports the practical objectives; supports the technological constraints and supports the educational directions for the longer term of the project.

5.0 Artefact Design

This chapter contains a factual account of the artifact developed for this project. The artifact is an Offline Speech Recognition Assisted Learning prototype to be developed for not only Chinese language learning but other languages. The design philosophy is built around the fact that there is a need to develop a localized and low latency means of allowing learners to improve their pronunciation skills, in addition to providing more privacy than what is available through most cloud-based learning systems.

The design philosophy is a combination of the original educational goals for the project and the changes that were made to the design as it progressed. At the start of the project, the intention was to develop a more comprehensive learning support system using a combination of speech recognition, pronunciation analysis, and visual feedback on errors made. As the project progressed, it became apparent that the most technically feasible and achievable element of the project within the specified time frame was to develop a working offline speech recognition prototype. For this reason, the design of the artifact reflects the use of locally loaded models, processing of speech into text format and user interface to provide an opportunity for expansion into more detailed pronunciation feedback functions later.

Thus, the artifact is designed as a building block for the development of further learning aids; and is not designed as simply a transcription tool.

5.1 Design Specifications

<i>Design Requirements</i>	<i>Specific instructions</i>
Run locally	The system supports offline speech recognition to reduce reliance on network connectivity.
Chinese recognition	The system is capable of basic recognition and text output for Chinese speech input.
Model controllable	Users can load the model and adjust inference-related parameters.
Complete interaction foundation	The interface includes recording control, recognition result display, and a reserved area for feedback presentation.
Scalability	Structural space is reserved for later expansion into pronunciation feedback, visual analysis, and mobile deployment.

Stability	The application maintains basic operational stability in the desktop development environment, and no serious functional errors were found during practical testing.
Privacy	The system is designed, as far as possible, to avoid uploading users' voice data and to protect learner privacy.

Table 1: Project design requirements and specifications.

The project requirements intertwined with the areas of concern as identified earlier in the project. Specifically, one area of concern identified during the problem analysis was the need for a system capable of supporting pronunciation development more practically and with less reliance upon third-party services. The downside of the offline-based implementation is that it would also preclude users of the system from taking additional advantage

The project has been developed while adhering to three underlying principles – offline execution, local control, and growth potential. Throughout the lifetime of the project these principles remained constant and as such, served as a guiding factor in the implementation of the system as well as in the evaluation process.

5.2 Artefact's Structure/ Architecture / Framework

The design of this project is divided into Four Modules; they are:

- 1) The Model Layer: It is the primary feature of the application. It operates as the core processing engine of the application and uses the ggml model formats for locally stored-use and allows inference to be done locally.
- 2) The System Integration Layer: It is responsible for the overall functionality of the application and controls the CMake build process. It also handles any dependencies needed for both the 'whisper' and 'ggml' libraries to function.
- 3) The User Interaction Layer: This module is where the end-user interacts with the application and consists of multiple features, including a button for loading models, controlling the audio recording process, displaying real-time transcription output, and providing feedback on the transcription results.
- 4) The Extension Layer: The extension layer of the application allows for future development of additional tools and features of the application in terms of pronunciation similarity scoring, providing highlighted error feedback and providing some level of gamification design to enhance learning. This design will serve as the basis for evolving from a basic speech recognition tool into a comprehensive pronunciation feedback system; the completion of this project

represents the first step of its development as a full-fledged Pronunciation Feedback System, which is an entirely new concept in developing speech-based learning applications. Future improvements to be included in the overall development process of this project will be the addition of teaching feedback functions.

6.0 Implementation

The process of building the artefact and how its core functionalities were validated through actual testing are described in this chapter. In contrast to the previous chapter's discussion of the design logic, systems structure and requirements, this chapter will provide a more practical focus on how the artefact was built.

The chapter explains how the artefact started as an idea and ultimately became a working offline prototype. The artefact's core technical choices were transformed into a working system via a series of steps that required coding, model integration, building configuration, constructing user interfaces, managing dependencies and extensive debugging until a stable version was created. Furthermore, it is important to place the development of this artefact within the context of its revised priorities. The original plan called for much greater integration of speech analysis, speech recognition and visual feedback, but the actual construction process focused primarily on developing a reliable offline speech recognition pipeline and user interface. This initial focus allowed the final artefact to meet the most critical technical aspect of the original project concept, while still permitting additional features to be added in the future.

6.1 Construction of the Artefact

The initial focus of the implementation process of the project was developing a working version of the core speech recognition engine for the offline use case. In relation to this, `whisper.cpp` was chosen as the primary engine during the comparison of solutions being proposed to provide the same functionality based on cloud-based APIs. This decision was made based on the proven reliability of the Whisper model when trained on large, multilingual, and multitasking datasets and its proven ability to generalise to multiple tasks associated with speech recognition (Radford, et al., 2023). Additionally, the decision to use `whisper.cpp` was appropriate for this project as it provides a lightweight implementation of Whisper in C/C++ and supports CPU-only inference and the ability to run with quantisation and multiple hardware backends (ggml-org, 2026). During this stage of the project, the system can now load models in the ggml format locally and perform basic speech-to-text processing of Chinese speech.

As the implementation of engineering for the project progressed, the CMake build system was the primary focus of the project. As part of the development cycle, the whisper and ggml source codes were gradually combined, which highlighted various compilation and linking issues that arose in the Visual Studio development environment. Several typical problems were also encountered during the engineering implementation stage.

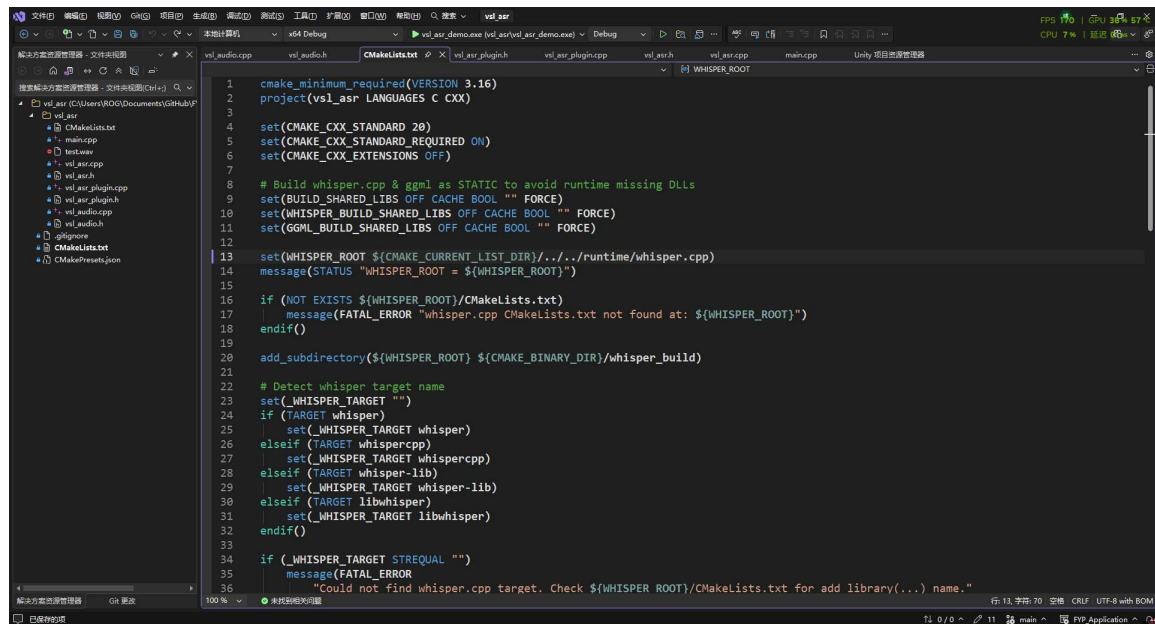


Figure 1: CMake configuration used for integrating whisper.cpp and ggml dependencies.

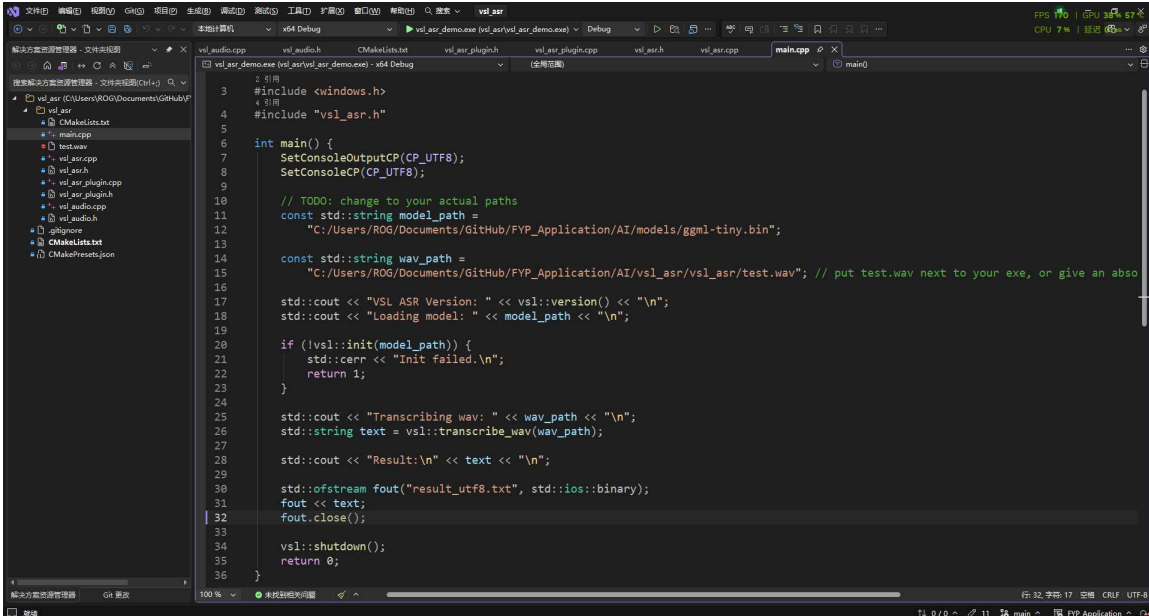
The first problem encountered was a `std::clamp` compilation error, which was attributed to a misconfigured compiler standard version on the machine. The second problem encountered was due to the development environment on Windows, where the path reading was incorrectly read, preventing the correct loading of models and associated resources. The third issue encountered was a GPU backend configuration conflict, which manifested in terms of an inconsistent acceleration performance across different environments. The fourth challenge was related to how complicated it is to manage dependencies and link the logic together, causing the project to continually adjust its build settings and code structure.

Through these developments, the project worked to improve the stabilisation of the whole system through the unification of compiler requirements, the reworking of portions of code logic, and the streamlining of both parameter configuration and pathing. Furthermore, these developments enable the project to illustrate that deploying an offline AI model within a live application creates a larger challenge than using the model alone; rather, the major challenge lies in finding

effective engineering methods for integrating the model into a well-functioning entire system.

Along with the main speech recognition pipeline, the project has created an elementary user interface framework. Presently, the model-loading button, controls for starting/stopping recording, and real-time transcription text display are the main functions that have been created. One area of the interface has also been set aside for when users provide feedback on their outcomes. Although it is important to note that the current interface is still being used primarily for verifying functionality, and the visual appearance of the interface has not yet been optimised, most of the key components of the system flow have been established. Hence, even at this early stage, the prototype has been established as capable of supporting functional testing upon the completion of its prototype stage and will be available for the later development of additional features.

Along with the Windows-based development and debugging environment, the project is also developing a basic Android version of the prototype. This is being accomplished through integrating the native speech recognition module into the Android project using both the Android NDK and the CMake build system. The existing Android project will enable the application to be able to call on the native local recognition module, which will run the core offline recognition workflow from a mobile device. The successful implementation of this element of the project has proven that the project is not only limited to desktop development but is also able to be transferred into the desired mobile learning setting.



```

1  #include <windows.h>
2  #include "vsl_asr.h"
3
4  int main() {
5      SetConsoleOutputCP(CP_UTF8);
6      SetConsoleCP(CP_UTF8);
7
8      // TODO: change to your actual paths
9      const std::string model_path =
10         "C:/Users/ROG/Documents/GitHub/FYP_Application/AI/models/ggml-tiny.bin";
11
12      const std::string wav_path =
13         "C:/Users/ROG/Documents/GitHub/FYP_Application/AI/vsl_asr/vsl_asr/test.wav"; // put test.wav next to your exe, or give an abso
14
15      std::cout << "VSL ASR Version: " << vsl::version() << "\n";
16      std::cout << "Loading model: " << model_path << "\n";
17
18      if (!vsl::init(model_path)) {
19          std::cerr << "Init failed.\n";
20          return 1;
21      }
22
23      std::cout << "Transcribing wav: " << wav_path << "\n";
24      std::string text = vsl::transcribe_wav(wav_path);
25
26      std::cout << "Result:\n" << text << "\n";
27
28      std::ofstream fout("result_utf8.txt", std::ios::binary);
29      fout << text;
30      fout.close();
31
32      vsl::shutdown();
33      return 0;
34  }

```

Figure 2: Core model loading and local speech recognition code.

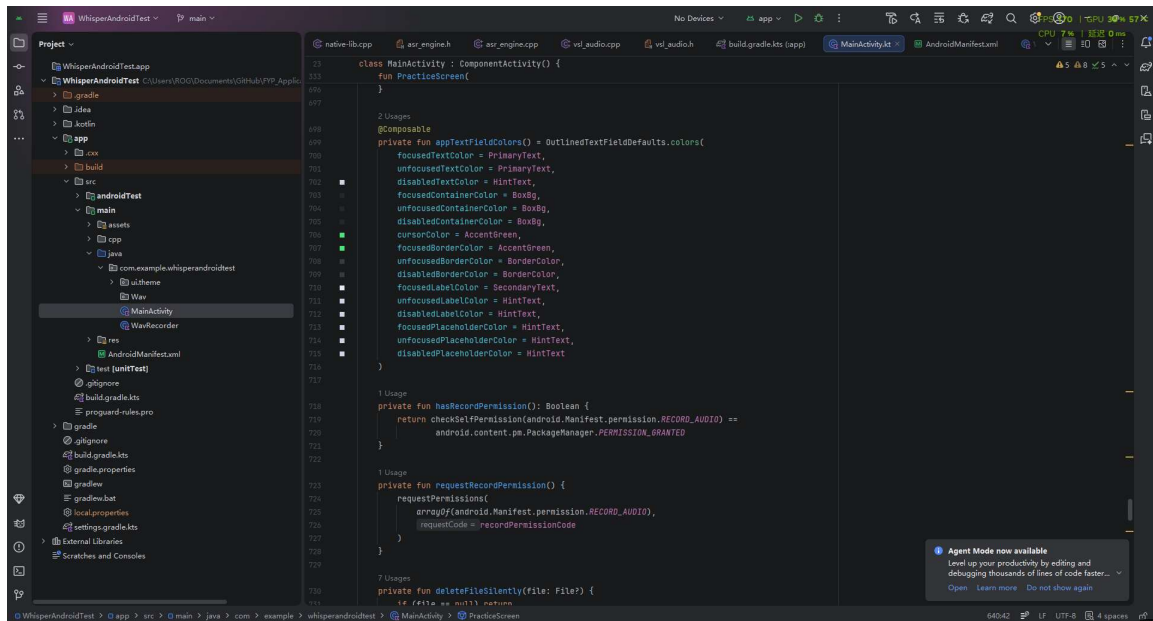


Figure 3: Android native interface used to connect the application with the local recognition model.



Figure 4: The real-machine interface on Android.

6.2 Validation and Testing

Based upon project progress thus far, system verification currently occurs at prototype functional level only vs. large scale experimental levels of experimentation. The verification outcomes at this stage were primarily evaluated based upon the following outcomes:

- 1) The ability for the system to successfully load locally stored models into memory and perform inference has demonstrated that the offline speech recognition approach was a feasible solution.
- 2) During basic functional testing, the system was able to provide stable performance when recognizing Chinese speech; this indicates there is a practical prototype that will perform the basic recognition tasks.
- 3) The performance stress test performed on the system with GPU acceleration enabled indicated several differences when compared against other device environments. When the GPU is enabled, inference speed increases; however, there are instances of anomalies in performance within certain device environments. Therefore, providing an option to configure some of the key parameters will improve the usability of the system.

In respect to privacy, the system is designed to process all input speech solely on the user's day-to-day use of the system, with no need for the upload of the user's voice to a remote cloud service. The audio recorded is temporary, used solely for the current recognition task, after which the recorded temporary audio will be deleted from the system. The prototype, therefore, performs no further processing of a user's speech beyond what is needed for recognition purposes. This design supports the project requirements for offline operation, lightweight operation and protection of personal privacy for the learner.

In addition, since the primary use of this system is for personal learning and not for public or broadly complex open usage, the system is highly sensitive to background noise and environmental noise. Initial testing showed that recognition accuracy is adversely affected by background noise, but the finding from informal functional testing of the prototype in the quiet environment demonstrated that the prototype produced usable results in the form of phrase-level Chinese transcription. Thus, the current prototype represents a prototype with some demonstrated utility, but further systematic testing using clearly defined test phrases in conjunction with multiple speakers and performance metrics is necessary to establish a stronger evidence base for the recognition performance of the system. Development of systems capable of noise-reduction, environmental robustness and adaptability to multi-scenario environments must

occur for the project to develop solutions that are viable in typical real-life settings.

The original project plan included a series of activities that included: performance stress-test, functional testing of 5-10 user participants and providing verification-of-pronunciation feedback; however, only the core-prototyped and verified functional feasibility have been completed as of the submission of this report. This report therefore considers this project to have achieved its core-prototyped and verified functional feasibility goal but does not elaborate any claims of completed implementation within a typically effective teaching experiment.

7.0 Critical Evaluation

7.1 Impact

The project has successfully developed from an idea to a functional prototype of a technology. The original project goal of creating a multimodal learning system has been expanded to include offline speech recognition, core system integration, and user interface. The prototype represents an advancement from the concept of a locally run AI for language learning to a working example of an AI.

This project has many strengths. The most significant is that the system can work without internet access, which provides the learner with additional privacy as well as independence from an internet connection. The system will also be effective for those learners in areas with unstable internet connectivity. Additionally, the development of this project did not view speech recognition as the end goal but rather a means of developing a more extensive pronunciation support system. The long-term research direction for this project is therefore clear and can be extended based on the speech recognition system. The project also has practical applications as it encountered and addressed real engineering challenges in building the model, configuring the build, and deploying the system, giving it both conceptual and practical contributions.

While demonstrating the development of this project, it has also been shown to be an essential difference between a technical prototype and a complete educational product. Although the current system can carry out offline speech-to-text processing, this is not enough to fulfil the greater needs of pronunciation learning.

8.0 Conclusion

An initial problem identified by this project was related to current pronunciation learning applications. These applications tend to provide low-quality, unclear, or insufficiently interpretable feedback to assist learners with improving their pronunciation skills. The main goal of this project was to determine if building a locally hosted offline speech recognition system could produce a better technical platform for providing pronunciation assistance than what is currently available. The project's initial phase included a development phase of early concepts. The project's ending result was not the same as what the original concept outlined, which included a more complete multi-modal system that included speech recognition, analysis and visual feedback. The project finalised an offline Chinese speech recognition prototype system with three main components: the ability to load local models, build and implement into interactive learning systems, and a basic user interface option. The limited length of time allowed for development is therefore in keeping with the evolving nature of the project. The completed prototype demonstrates that offline speech recognition technology can now be successfully incorporated into a learning-based system running on personal devices. Additionally, the prototype demonstrates that an offline learning-based speech recognition system can provide privacy protection, local control, and independence from any third-party hosted cloud services. As a result of the implementation and development of this project, it was further clarified that the challenges in making offline AI practical for learning-based applications relate not only to the selection of models and data in creating speech recognisers, but also to the challenge of efficiently and correctly developing the software, including issues related to compiler, dependency and path handling, setting up the GPU for machine learning, and developing an easy-to-use software interface. Overall, the results of this project demonstrate that it has successfully completed its technical goal of taking a conceptualisation of an idea and creating it into a working prototype that will provide a basis for further educational development of pronunciation-based learning tools. While currently, the prototype does not provide all of the features necessary for learners to receive complete and clear feedback on their pronunciation, the framework outlined by the prototype will support all of the key components that must eventually be developed and included in future systems, including the ability to identify specific pronunciation errors, provide similarity scores between the speaker's and the reference pronunciations, and produce more clearly interpretable guidance for learners on how to improve their pronunciation skills.

8.1 Limitations

Though the ugly prototype phase is complete, there are still limits to the system. First, the current system primarily demonstrates an offline speech recognition algorithm and basic interface functionality and still does not demonstrate giving a complete pronunciation feedback loop, so educationally speaking at this stage it is the groundwork for a teaching or learning tool but not the tool itself. Second, although we built a rough prototype for Android, too, and have living prototype versions running on Windows and Android devices, the cross-device testing is relatively sparse. The Android version currently demonstrates that our offline recognition flow will run on a mobile device, but a lot of things could still be optimised: loading speed of model files, memory footprint, responsiveness of the interface, and making it work across various Android phones, for example. Third, though the original project plan called for inviting 5, 10, or more users for functional and usability testing to try out, we are not quite to that level of systematic round or proof of experimental data yet, and so we are not able to claim much in the way of educational testing results in this report. Finally, as our prototype currently validates is mostly a phrase-level Chinese speech recognizer, its support for continuous speech + more complex learning scenarios, etc, and its richer types of language and speech learning situations are still somewhat limited.

8.2 Future Work

Future work can proceed in the two directions:

First, further development of the pronunciation feedback mechanism. As discussed in the later phases of the project plan, this is the real educational substance behind the system, and future work can aim to implement syllable comparisons and feedback, similarity scores for the comparison, marking errors on-screen, and finally organizing it all into a structured user feedback output.

The next step is to optimise the system's performance through stress testing and identify the maximum number of users a given configuration can handle before it fails. Further development of future improvements should concentrate on providing users with a fast and responsive recognition solution that performs consistently across devices and will continue to function well after multiple usages.

The final step is to increase the number of participants in the user testing process and evaluate usability testing. Additional participants from China could be recruited to provide a larger database of user experience information as it relates to the recognition system's performance, the appropriateness of feedback

provided on the screen, the speed at which users learn the recognition system, and overall user satisfaction with the tool. This will increase the effect each of those issues have in the project.

Future work could also expand the learning content to include richer vocabulary, everyday expressions, and lightweight gamification.

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Appendices

A.1 Project Timeline

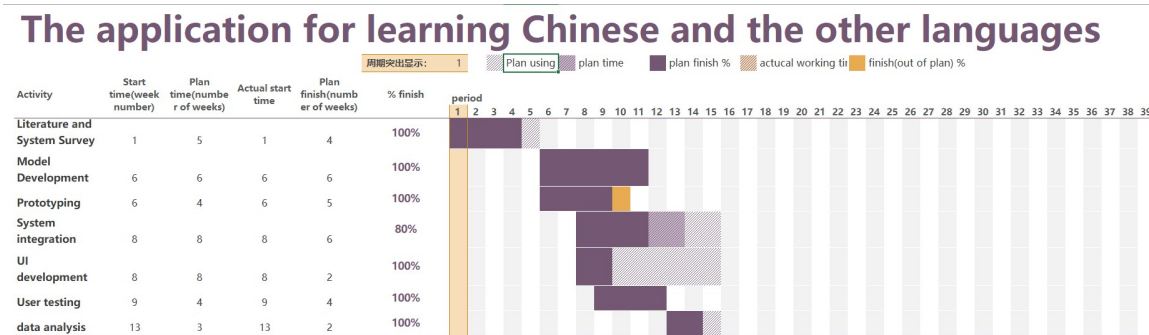


Figure 5: Project timeline and Gantt chart.

The Gantt chart above indicates the main stages of the project: ‘literature review’, ‘prototype design’, ‘Windows prototype development’, ‘Android implementation’, ‘functional testing’, ‘questionnaire feedback’, ‘final report writing’.

A.2 Feasibility, Risks, and Ethical Aspects

This section summarises the practicality, risks, and ethical considerations of the project. Since the project was scaled down from a fully multimodal pronunciation-feedback system and interpreted instead as an offline recognition-assisted prototype, the scope that was achieved was realistic with time and available technical and resources. The practical impact was aimed at creating a working Windows prototype with the current technology available and moving to a more basic implementation on Android. The “local” speech recognition was validated, and the future consider how the system could be expanded to become pronunciation feedback.

B.1 Revised Feasibility

While the project was still broadly feasible in the time available, it simplified to a prototype rather than an application, seeking out and utilizing resources readily available within this timeframe - a windows development environment, Android Studio, CMake, the Android NDK, whisper.cpp and ggml-format speech recognition models. While the concept initially included speech recognition, tone analysis and visual feedback, the project managers chose to enable simpler offline recognition pipeline first and then the basics of an android application. Then, despite simplifying the scope, left with a very clear base for pronunciation-feedback functions.

B.2 Updated Risk Analysis and Mitigation

Potential Risk	Likelihood (Low/Med/High)	Impact (Low/Med/High)	Mitigation Strategy
<i>Difficulty integrating whisper.cpp with Android</i>	<i>Medium</i>	<i>High</i>	<i>Use Android NDK and CMake step-by-step and test the native interface separately.</i>
<i>Mobile device performance</i>	<i>Medium</i>	<i>High</i>	<i>Use lightweight ggml models and optimise model loading.</i>
<i>Recognition affected by noise</i>	<i>Medium</i>	<i>Medium</i>	<i>Test in quiet environments and identify noise reduction as future work.</i>
<i>Limited user Testing sample</i>	<i>High</i>	<i>Medium</i>	<i>Use questionnaire feedback only as preliminary evidence</i>

Table 2: Risk Assessment and Mitigation

B.3. Revised Ethical Considerations

This project studied ethical implications of the handling of speech data, feedback from questionnaires and learner privacy. The prototype speech input would be processed locally instead of uploading user voice data to the cloud. Any recorded speech that was used during testing was only used to verify functionality and was not kept for different types of analysis afterwards. Questionnaires were used only for initial feedback on learner expectations and did not ask participant for

personal identifying information. Participation was voluntary, and results again were feedback, and not to be considered statistically generalizable.

Appendix C: Questionnaire Results

Below, a summary a small-scale questionnaire that this project used to garner some early feedback on learner expectations of a Chinese pronunciation-support application. As it was anonymous, the project used the info closed to be able to contextualise feedback used elsewhere in the project. There were responses.

What part you think is the most difficult partt for you?

(9 条回复)

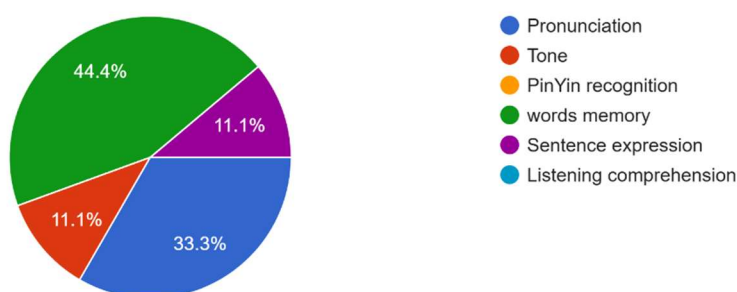


Figure 6: Main difficulties reported by respondents when learning Chinese.

The result shows which aspects of Chinese learning respondents considered most difficult. This provides background support for the project's focus on pronunciation and tone-related learning problems.

Do you think the "Real-time pronunciation correction" is important during learning chinese?

(9 条回复)

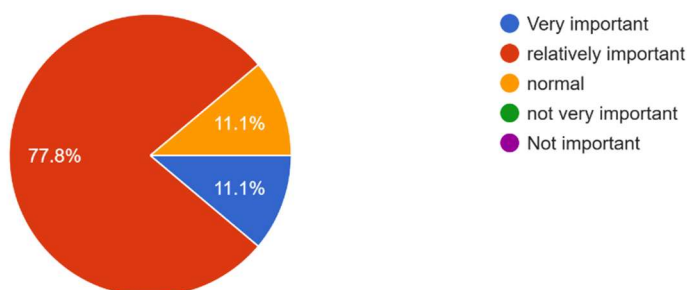


Figure 7: Respondents' views on the importance of real-time pronunciation correction.

The result indicates whether respondents considered real-time pronunciation correction important during Chinese learning. This supports the project's focus on immediate feedback as a useful direction for pronunciation-support systems.

What problems do you think exist in current Chinese learning software regarding pronunciation practice? (Multiple selections allowed)

(9 条回复)

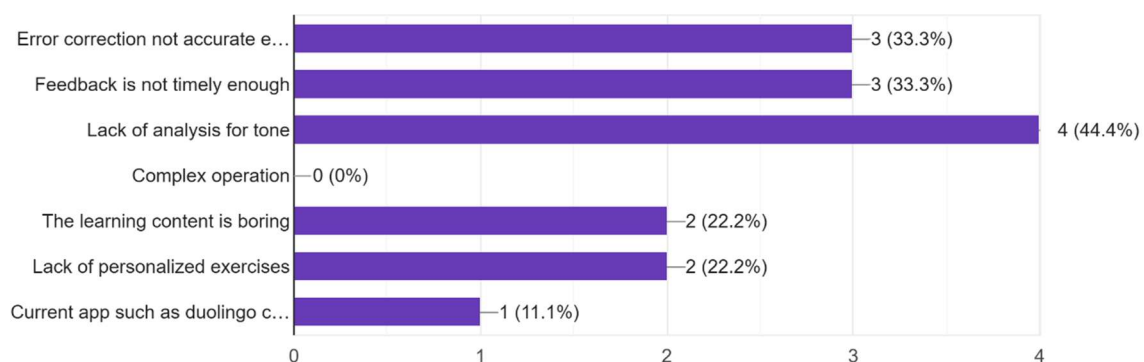


Figure 8: Problems identified by respondents in existing Chinese learning applications for pronunciation practice.

The responses highlight perceived weaknesses in existing Chinese learning applications, such as limited tone analysis, inaccurate correction, delayed feedback, or insufficient personalised support. These results support the problem statement of this project.

What features would you most like this Chinese learning app to have? (Multiple selections allowed)

(9 条回复)

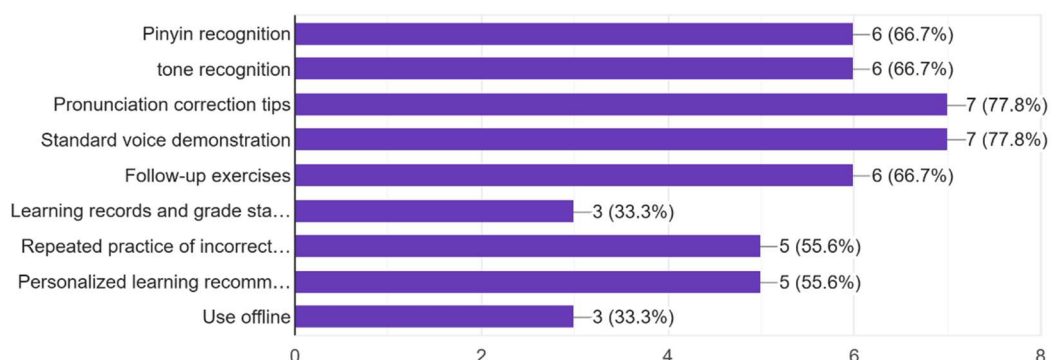


Figure 9: Features that respondents would most like the proposed Chinese learning application to include.

The result shows which functions respondents expected from the proposed application. Commonly selected features can be used to justify future development directions such as pronunciation correction, tone feedback, standard pronunciation examples, and offline usability.

If the app can perform basic speech recognition and error correction without an internet connection, would that increase your willingness to use it?

(9 条回复)

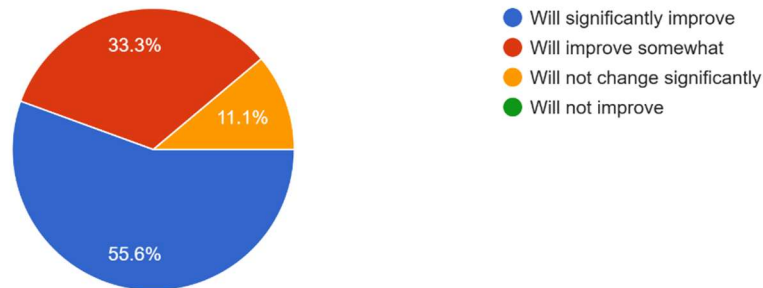


Figure 10: Respondents' willingness to use the application if offline speech recognition and correction were available.

The result is directly relevant to the offline design of the prototype. It indicates whether local speech recognition and reduced reliance on internet connectivity could increase respondents' willingness to use the application.